

Neel Gajare

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. EECS (Electrical Engineering & Computer Science)

May 2027

Relevant Coursework (3.85 GPA): Advanced Algorithms, Computer Architecture, Machine Learning, Data Structures, Optimization Models in Eng, Circuit Design, Discrete Math & Probability, Linear Algebra, Multivariable Calculus, Electricity & Magnetism, Mechanics, Competitive Programming, Digital Design and Integrated Circuits, FPGA Lab, Advanced LLM Agents
Societies: Space Technology @ Cal, Cloud Computing @ Cal, Codify

EXPERIENCE

SWE Intern @ Amazon

May 2025 – Present

Amazon

Sunnyvale, CA

- Designing low-latency traffic analysis workflows to detect and identify unique non-volumetric and agentic bot signatures affecting Amazon's retail websites and backend services.
- Utilizing AWS tools such as Athena, Sagemaker AI, Cloudwatch, EC2, and internal tools to automate manual traffic spike investigation time for oncall engineers by >90% and reduce compute costs by >85% on average.

Berkeley AI Research

Feb 2024 – May 2025

Berkeley AI Research Lab (BAIR)

Berkeley, CA

- Worked under Professor Kurt Keutzer on Generative AI project pertaining to low resource machine translation and linguistic segmentation.
- Utilized Langchain, Chroma, Milvus to implement efficient RAG techniques on billion-token corpus to improve translational model. Integrating vector database to backend system and API with FastAPI. Implemented agentic workflows for paragraph segmentation techniques to enhance retrieval system.

Stanford Research Intern

May 2022 – Feb 2024

Stanford University

Palo Alto, CA

- Worked in Stanford's Crustal Deformation Lab under Professor Paul Segall. Researched complex basaltic magma chamber geometries using AI.
- 1st Author of abstract published in American Geophysical Union (AGU) conference proceedings, presented in AGU Fall Meeting.
- Utilized Gmsh for 3-D mesh generation to model magma chambers, Matlab for simulation of surface deformation and Bayesian optimization techniques.

AWARDS

- Apple FPGA Digital Design Contest - 1st Place May 2025
- NVIDIA National Merit Scholarship April 2023
- St. Francis de Assisi Volunteering Award May 2022

PROJECTS

RISC-V CPU on FPGA | *Verilog*

Jan 2025 – May 2025

- Implemented 5-stage pipelined RISC-V CPU on FPGA, with UART tethering, BIOS functionality, synchronous memories.
- Optimized CPU to run on 125 MHz clock frequency and 1.06 average CPI on various benchmarks by resolving pipelining hazards, designing a pipelined hybrid Gshare branch predictor + branch target buffer + return address stack.
- Achieved highest figure-of-merit score in Berkeley's digital design competition sponsored by Apple.

IntrospectAI | *React, Three.js, Flask, Claude, Supabase*

Jun 2025

- Built real-time 3D visualization tool that transforms ChatGPT and Claude conversation histories into interactive, AI-labeled semantic clusters using Voyage embeddings and Claude Sonnet 4 for topic detection and RAG.
- Created a high-performance React + Three.js frontend with GPU-accelerated WebGL shaders and a Flask backend supporting multi-user interaction, live API updates, semantic search (pgvector), and dynamic clustering.
- Designed a resilient NLP pipeline with SpaCy, UMAP, KMeans, and Claude for auto-labeling; supported cross-platform chat parsing, mock data fallback, and smooth UX for filtering, zooming, and exploration of AI chat patterns.

UniTrial | *Python, Firebase, Streamlit, HuggingFace, ChromaDB, Mistral AI*

Jun 2024

- Developed RAG system to match patient EHR profiles/ medical condition descriptions with clinical trials using Mistral-7B.
- Implemented novel data schema leveraging FHIR and MESH IDs to log medically relevant tags, preventing hallucinations from unstructured data.
- Created RAG pipeline with chat interface for users with to ask questions about trial & get diagnosis.

TIME II | *STM32 Microcontroller, PCBA*

Jan 2024 – Present

- Working on project at Space Technology @ Cal, which will send a payload to a suborbital Blue Origin flight to conduct E. coli microgravity experiments.
- Programmed STM32 microcontroller to maintain optimal temperature in the capsule via thermistors.

TECHNICAL SKILLS

Languages: Python, Java, Verilog, Matlab, SQL, HTML/CSS, Scheme, Javascript, C, C++, RISC-V Assembly

Frameworks/Tools: AWS EC2, Athena, Sagemaker AI, Trainium, LangChain, Pytorch, Vivado, Keras, OpenCV, Gmsh, Tableau, Numpy, JUnit, Arduino, Node.js, Express, React, Tensorflow, MongoDB, OpenAI, HuggingFace, Chroma, FastAPI, Firebase, Valgrind, Streamlit, Milvus, GDB, TensorRT

Skills: AI/ML, Chip Design, Computer Architecture, Algorithms, Data Processing/Analysis, PCB Assembly, NLP